Class Activity # 5
Calculus Physics 2

ame: Key Date: _____

Group Members: _____ _____ _____

Solve the following problems together as a group, stay engaged in the conversation with the group!

- Use **Lecture Notes Chapter 48 through 52** as a guide to answer the following questions.

1) Magnetic flux through a loop can change even when the magnetic field is constant because:

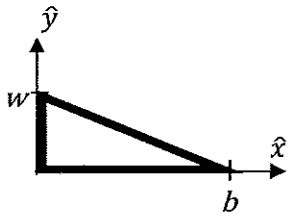
A) Flux does not depend on geometry
B) The loop can rotate or change area
C) The loop generates its own magnetic field
D) Flux is independent of the angle between the loop and field

2) A student claims: "There is no induced emf if the magnetic field is constant." Why is this incorrect?

A) The student forgot about Lenz's law
B) A changing loop area can change magnetic flux
C) Emf is unrelated to flux
D) Constant fields cannot exist

3) A hollow wire triangle with the lengths of the sides b and w sits in a magnetic field with a x dependence. Determine the equation for the magnetic flux through the triangle for a magnetic field that changes in the x direction given by the equation below were C is a constant.

$\vec{B} = (Cx)\hat{z}$



see attached

$\Phi_m = \frac{Cwb^2}{6}$

4) The magnetic field strength is changing with time as shown in the equation below inside a coil of wire with an area of 2 x 10^{-3} m^2 and 100 loops of wire. The normal to the surface area of the coil make an angle of 30° to the magnetic field lines. Determine the induced potential at t = 5 s given that

$\vec{B} = [(3T/s)t + (0.5\ T/s^2)t^2]\hat{z}$

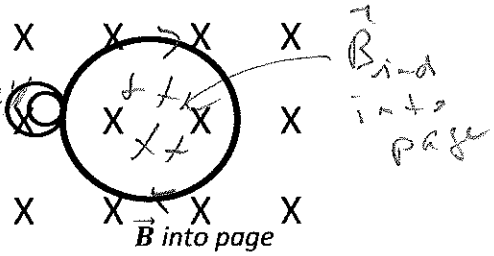
$V_{emf} = -N\frac{d\Phi}{dt} = -N\frac{d}{dt}\left(\int \vec{B}\cdot\hat{n}\,dA\right) = -NA\left(\frac{dB}{dt}\right)\cos\phi$

$V_{emf} = -NA[3\,T/s + 1\,T/s\ t]\cos\phi = 1.4\,V$

5) A wire loop (N = 1) with a light bulb (R = 25 Ω) with an area of 0.3 m^2 is placed in a 0.5 T uniform magnetic field $\vec{B}$ that is pointing into the page, if the magnetic field drops from 0.5 T to 0 T in a time of 2 x 10^{-3} s.

A) What direction is the current be induced?

RHR #2 → decreasing B_{ext} → Induced field adds to B_{ext}



B) Determine the current induced in the loop?

$V_{ind} = -N\frac{\Delta\Phi}{\Delta t} = -NA\frac{(B_f - B_i)}{\Delta t} = 75V$

$I = \frac{V_{ind}}{R} = 3A$

C) What would be the direction of the induced magnetic field and the magnitude of the induced current if the field is turned on and the external field changes from 0 T to 0.5 T in 2 x 10^{-3} s?

$V_{ind} = -N\frac{\Delta\Phi}{\Delta t} = -NA(B_f - B_i) = -75V$

$I = \frac{V_{ind}}{R} = -3A \rightarrow CCW$

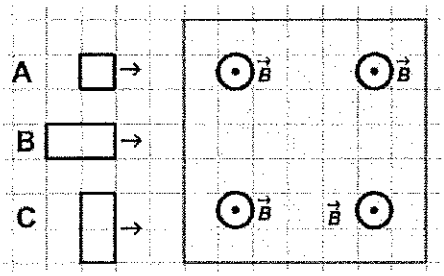
D) What if the magnetic field was held at a constant 0.5 T but the wire loop was rotated from 0° to 90° in 2 x 10^{-3} s, what would be the induced current in the wire?

$V_{ind} = -N\frac{\Delta\Phi}{\Delta t} = -N\frac{AB(\cos(\phi_f) - \cos(\phi_i))}{\Delta t} = 75V$

6) The figure shows three metal coils labeled A, B, and C heading towards a region where a uniform static magnetic field exists. The coils move with the same constant velocity and all have the same resistance. Their relative sizes are indicated by the background grid. As they enter the magnetic field the coils will have an induced electric current in them. For which coil will the current be the greatest?

A) A
B) B
C) C
D) The current is the same in all three cases since all the coils move with the same velocity.
E) There is no induced current in any of the coils since they move at constant velocity.

$V_{ind} = BvL$ ← length in direction of $\vec{v}$

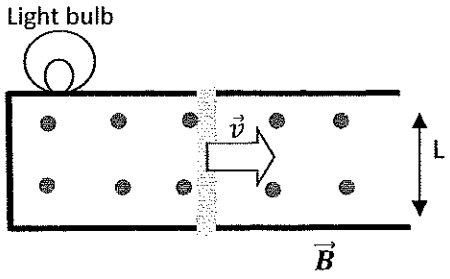


11) A conducing rod is moving perpendicular to a uniform magnetic field ($|\vec{B}|$ = 0.5 T) that is directed out of the page and an induced current is created. The rod is traveling on a frictionless conducing rail that are separated by L = 1.5 m, with a light bulb that has a resistance of R = 6 Ω. The rod travels at a constant rate of 20 m/s, what is the force required to pull the rod at the constant velocity of 20 m/s?

$V_{ind} = BvL\cos\phi = 15V$

$I = \frac{V}{R} = \frac{15V}{6\Omega} = 2.5A$

$|\vec{F}_m| = ILB = 1.875N$



11) A battery charger uses mutual induction (Transformer) to charge a flashlight; the charger is a solenoid made up of a $N_p = 500$ loops of wire at $V_p = 120$ V. The coil inside the flashlight needs 6 V to charge the batteries. Assume the charger and flashlight act as a transformer when charging,

A) Determine the number of loops of wire needed for the coil inside the flashlight.

$\frac{N_s}{N_p} = \frac{V_s}{V_p}$ | $N_s = N_p\left(\frac{V_s}{V_p}\right) = 25$ loops

B) If the batteries in the flashlight need 4 W to charge properly, determine the current in the chargers' coils (primary coil) as the batteries charge.

$\frac{I_p}{I_s} = \frac{V_s}{V_p}$ | $P_s = I_s V_s$, $I_s = \frac{P_s}{V_s}$ | $I_p = I_s\left(\frac{V_s}{V_p}\right) = \frac{P_s}{V_s}\left(\frac{V_s}{V_p}\right)$, $I_p = \frac{P_s}{V_p} = 0.033\,A$

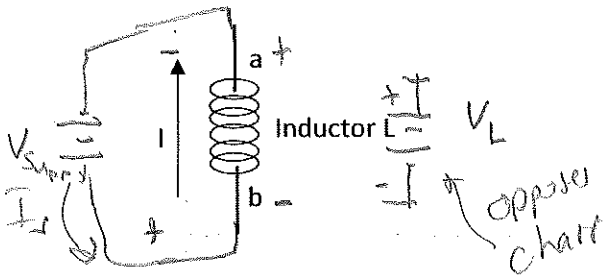
C) Determine the amount of power required by the primary coils.

$P_p = I_p V_p = 4\,W$

12) An Inductor has current flowing through it in the direction shown. If the current is getting larger with time, what polarity are the positions on either side of the inductor marked "a" and "b".?

A) a is positive and b is also positive.
B) a is negative and b is also negative.
C) a is positive and b is negative.
D) a is negative and b is positive.
E) There is no induced polarity in the inductor.

(C)



13) An inductor has a time dependent current flowing through it shown in the equation below. If the maximum emf across the inductor is equal to 0.5 V, and the angular velocity $\omega = 314$ /s, determine the self-inductance (L) of the inductor?

$I(t) = (0.5\,A)\cos(\omega t) = 0.5A\cos([314/s]t)$

$V_{ind} = -L\frac{dI}{dt}$

$\frac{dI}{dt} = (0.5A)(314/s)\sin(\omega t)$, $\sin(\omega t)_{max} = 1$

$L = \frac{V_{ind}}{(0.5A)(314/s)} = 0.32\,H$

14) An electrical generator on a bike is wired to a light bulb ($R = 4\ \Omega$). The generator armature is made up of 500 loops of wire, has an area of $2 \times 10^{-4}\ m^2$ and is in a 0.05 T magnetic field.

A) Determine the frequency the generator must run to generate an average power of 6 W?

$V_{rms} = \frac{NAB(2\pi f)}{\sqrt{2}}$ | $P = \frac{V_{rms}^2}{R}$ | $f = \frac{\sqrt{2}\,V_{rms}}{NAB2\pi}$

$P = IV = V^2/R$

$V_{rms} = \sqrt{PR} = 4.9V$ | $f = 220.5\,Hz$

B) If the radius of the sprocket that rests on the bike tire and turns the armature is r = 0.5 cm, at what velocity must the bike be traveling to generate 5 V_{rms}? Is this velocity reasonable? (remember the tangential velocity $v_T = 2\pi fr$)

$v_T = 2\pi f r = 6.93\,m/s \approx 14\,mph$

12) Radio Frequency Identification (RFID) chip is used to indicate that a runner has crossed a particular station in the race. At a station a changing magnetic field is created at 900 MHz (Radio wave), this signal creates a changing magnetic field from 0 T to 9×10^{-8} T in 4×10^{-10} seconds, if the RFID chip has 8 coils of wire with an area of 0.002 m^2 and a resistance of R = 4 Ω, what is the electrical power created in the chip by the signal?

$V_{ind} = -N\frac{\Delta\Phi}{\Delta t} = -\frac{NA(B_f - B_i)}{\Delta t} = 3.6V$

$I = \frac{V_{ind}}{R} = 0.9A \qquad P = IV = 3.24W$

15) In an RCL circuit describe each element's phase angle between the current and voltage, how it relates to the current through the resistor and how does the voltage across the element change as the frequency of the ac source is increased.

	Phase between Vand I	Relates to resistor	Voltage as f is increased
Resistor:	0°	leads or lags, same	depends on current
Inductor:	90° in front ELI	leads or lags, same	increases
Capacitor:	90° behind ICE	leads or lags, same	decreases

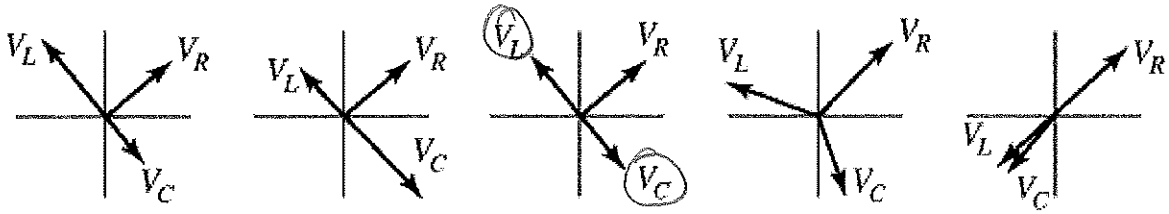
16) When an *LRC* series circuit is driven at resonance, which of the following statements about the circuit are correct? (There is more than one correct choice.) (C) (E) (I)

$X_L = X_C \qquad Z = R$

A) The impedance of the circuit is zero.
B) The impedance of the circuit has its maximum value.
(C) The impedance of the circuit has its minimum value.
D) The inductive reactance and the capacitive reactance are both zero.
(E) The inductive reactance and the capacitive reactance are exactly equal to each other.
F) The reactance of the inductor is zero.
G) The reactance of the capacitor is zero.
H) The reactance due to the inductor and capacitor has its maximum value.
(I) The current amplitude is a maximum.

17) Which one of the phasor diagrams shown below best represents a series *LRC* circuit driven at resonance?

(3)



1 2 (3) Equal 4 5

18) At what frequency does a 10 μF capacitor have a reactance of 1200 Ω?

$X_C = \frac{1}{2\pi f C} \Rightarrow f = \frac{1}{2\pi C X_C} = 13.3\,Hz$

19) An ac circuit has a 100 Ω resistor in series with a 4.9 μF capacitor and a 700 mH inductor. At what frequency does the circuit act like a pure resistance circuit?

$f = \frac{1}{2\pi\sqrt{RC}} \Rightarrow$ $Z = R$ at resonance | $f = 86\ Hz$

20) A series *RLC* circuit has a 100 Ω resistor, a 1 μF capacitor and a 2.0 mH inductor connected across a 120 V rms ac power source.

A) Determine the voltage across the resistor when the ac power supply's frequency is tuned to 1,000 Hz.

$X_L = 2\pi f L = 12.6\ \Omega$

$X_C = \frac{1}{2\pi C f} = 159\ \Omega$

$R = 100\ \Omega$

$Z = \sqrt{R^2 + (X_L - X_C)^2} = 170.4\ \Omega$

$I = \frac{V}{Z} = 0.68\ A$ through Resistor

$V_R = IR$

$V_R = 68\ V$

B) Determine the average power used by the circuit at 1,000 Hz

$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = 55.7^\circ$

$P = IV\cos\phi = 46\ W$

C) Determine the rms value of the current through the resistor when the ac power supply's frequency is tuned to the resonate frequency.

$f = \frac{1}{2\pi\sqrt{LC}} = 3558.8\ Hz$

$X_L = 44.7\ \Omega$

$X_C = 44.7\ \Omega$

$X_L = X_C$

$Z = R$

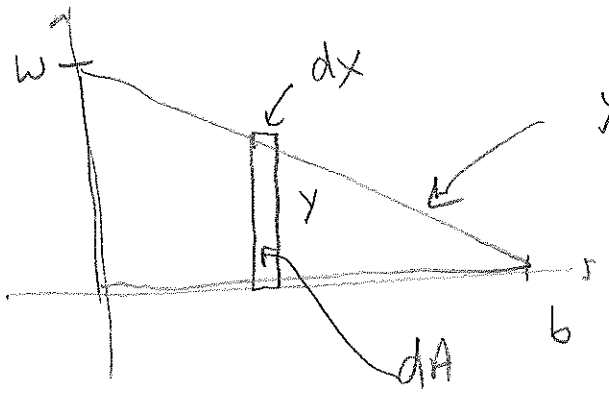
$I = \frac{V}{Z} = \frac{V}{R} = 1.2\ A$

D) Determine the average power used by the circuit at resonance.

$\phi = \tan^{-1}(0) = 0^\circ$

$P = IV = 144\ W$

3)



$$y = -\frac{w}{b}x + w$$

$$dA = y\,dx$$

$$dA = \left(-\frac{w}{b}x + w\right)dx$$

$$\Phi = \int \vec{B}\cdot d\vec{A}$$

$$\Phi = \int_0^b \left[(cx)\hat{z}\cdot\hat{n}\left(-\frac{w}{b}x + w\right)\right]dx$$

$$\Phi = cw\int_0^b \left[\left(-\frac{x^2}{b}\right) + x\right]dx$$

$$\Phi = cw\left[-\frac{x^3}{3b} + \frac{1}{2}x^2\right]_0^b$$

$$\Phi = cw\left[-\frac{b^{3^2}}{3b} + \frac{1}{2}b^2\right]$$

$$\Phi = cw\left[\frac{b^2}{6}\right] = \boxed{\frac{cwb^2}{6}}$$